Midterm 2

Name: Solution

Student ID:

By signing below, you affirm that you have neither given nor received unauthorized help on this exam.

Signature:

DO NOT OPEN THE EXAM UNTIL YOU ARE TOLD TO DO SO! YOU WILL GET A ZERO IF YOU DO.

This exam contains 5 pages (including this cover page) and 6 problems. The exam will be graded out of 100 points. **This is an individual exam.** You are not allowed to use a calculator, any additional notes, look at anyone else's paper, have extra paper, talk to anyone, consult the internet, anything. It is considered cheating if you use additional notes, have wandering eyes, are caught talking, etc. Absolutely no headphones, smartwatches, or cellphones out! **I have a zero tolerance policy for cheating.**

Instructions: Print your name or student ID number above AND ON EVERY PAGE. You may use any available space on the exam for scratch work.

Good Luck!

To Receive Full Credit

- **Answer must be written in the box provided!** Show your work on the rest of the page.
- You must **show your work** to receive credit.
- **Proper Notation and Organized.** Clear, concise, and correct use of mathematical notation is part of a correct answer. Work should be organized in a reasonable neat and coherent way in the space provided. Work scattered all over the page without a clearing ordering is not acceptable.

Do not write in the table to the right.

Problem	Points	Score
1	5	
2	10	
3	25	
4	15	
5	25	
6	20	
Total:	100	

1. (5 points) Consider the function $f(x) = x^3\cos(2x)$ defined on the interval $[-L, L]$. Is this function even, odd, or neither? Based on your answer, which of its Fourier coefficients (a_0, a_n, b_n) must be zero? (No calculation required).

$f(-x) = (-x)^3 \cos(-2x)$
$= -x^3\cos 2x$

not necessary

$f(x)$ is odd $\Rightarrow a_0 = 0,\ a_n = 0,\ b_n \neq 0$

---

2. (10 points) Consider the ODE $y'' + \lambda y = 0$ defined on the interval $[0, 2]$ with $y'(0) = 0$ and $y(2) = 0$. Determine the corresponding eigenvalues and eigenfunctions.

$y(x) = A\cos(\sqrt{\lambda}\,x) + B\sin(\sqrt{\lambda}\,x)$

$y'(x) = -A\sqrt{\lambda}\sin(\sqrt{\lambda}\,x) + B\sqrt{\lambda}\cos(\sqrt{\lambda}\,x)$

$y'(0) = B\sqrt{\lambda} = 0 \Rightarrow B = 0$

$y(2) = A\cos(2\sqrt{\lambda}) = 0$

$\cos(2\sqrt{\lambda}) = 0$

$2\sqrt{\lambda} = \frac{(2n+1)\pi}{2}$ or $2n-1$ w/ $n = 1, 2, 3, \ldots$

$$\boxed{\lambda = \frac{(2n+1)^2\pi^2}{4^2},\ n = 0, 1, 2, \ldots \qquad y(x) = \cos\left(\frac{(2n+1)\pi x}{4}\right)}$$

3. Consider $f(x) = \begin{cases} x+1 & -2 \le x < 0 \\ 2 & 0 \le x < 2 \end{cases}$

(a) (15 points) Compute the Fourier Series for $f(x)$. You may stop simplifying the expression for coefficients when there is no more sines or cosines.

$$\boxed{f(x) = 1 + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2}(1-(-1)^n)\cos\left(\frac{n\pi x}{2}\right) + \frac{(1-3(-1)^n)}{n\pi}\sin\left(\frac{n\pi x}{2}\right)}$$

$$a_0 = \frac{1}{2}\left[\int_{-2}^{0} x+1\,dx + \int_0^2 2\,dx\right] = \frac{1}{2}\left(\frac{1}{2}x^2+x\right)\Big|_{-2}^{0} + x\Big|_0^2 = \frac{1}{2}\left(\frac{1}{2}(4)-2\right)+2$$

$$= 1-1+2 = 2$$

$$a_n = \frac{1}{2}\int_{-2}^{0}(x+1)\cos\left(\frac{n\pi x}{2}\right)dx + \int_0^2 \cos\left(\frac{n\pi x}{2}\right)dx$$

$$= \frac{1}{2}\left(\cancel{\frac{2(x+1)}{n\pi}\sin\left(\frac{n\pi x}{2}\right)}\Big|_{-2}^{0}\right) - \frac{1}{n\pi}\int_0^2 \sin\left(\frac{n\pi x}{2}\right)dx + \left(\cancel{\frac{2}{n\pi}\sin\left(\frac{n\pi x}{2}\right)}\Big|_0^2\right)$$

$$= \frac{2}{n^2\pi^2}\cos\left(\frac{n\pi x}{2}\right)\Big|_{-2}^{0} = \frac{2}{n^2\pi^2}\left(1-(-1)^n\right)$$ don't need it simplified further

$$b_n = \frac{1}{2}\int_{-2}^{0}(x+1)\sin\left(\frac{n\pi x}{2}\right)dx + \int_0^2 \sin\left(\frac{n\pi x}{2}\right)dx$$

$$= \frac{-(x+1)}{n\pi}\cos\left(\frac{n\pi x}{2}\right)\Big|_{-2}^{0} + \frac{1}{n\pi}\cancel{\int_{-2}^{0}\cos\left(\frac{n\pi x}{2}\right)dx} - \frac{2}{n\pi}\cos\left(\frac{n\pi x}{2}\right)\Big|_0^2$$

$$= \frac{-1}{n\pi} - \frac{1}{n\pi}\cos(n\pi) - \frac{2}{n\pi}\cos(n\pi) + \frac{2}{n\pi}$$

$$= \frac{1}{n\pi} - \frac{3}{n\pi}(-1)^n$$

(b) (5 points) What value does the Fourier Series for $f(x)$ converge to at $x = 4$?

$$\frac{1}{2}\left(f(4^-)+f(4^+)\right)$$

$$= \frac{1}{2}(1+2)$$

$\boxed{3/2}$

(c) (5 points) What value does the Fourier Series for $f(x)$ converge to at $x = -1$?

$f(-1) = 0$

$\boxed{0}$

4. Consider the function $f(x) = x$ defined on $[0, \pi]$.

(a) (10 points) Find the **Fourier Cosine** series for $f(x)$.

$$\boxed{\frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2}\left[(-1)^n - 1\right]\cos(nx)}$$

$$a_0 = \frac{2}{\pi}\int_0^\pi x\,dx = \frac{2}{\pi}\left(\frac{1}{2}x^2\right)\Big|_0^\pi = \pi$$

$$a_n = \frac{2}{\pi}\int_0^\pi x\cos(nx)\,dx$$

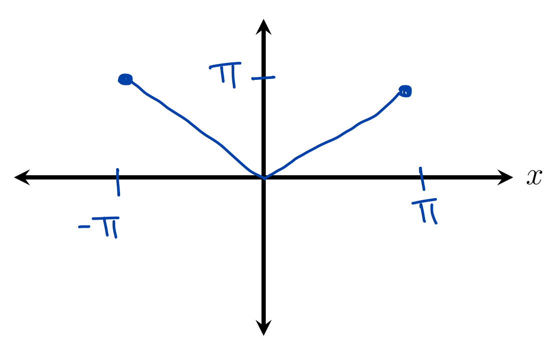
$$= \frac{2}{\pi}\left[\frac{x}{n}\sin(nx)\Big|_0^\pi - \frac{1}{n}\int_0^\pi \sin(nx)\,dx\right]$$

$$= \frac{2}{\pi}\left(\frac{1}{n^2}\cos(nx)\Big|_0^\pi\right)$$

$$= \frac{2}{\pi n^2}\left(\cos(n\pi) - \cos(0)\right)$$

$$= \frac{2}{\pi n^2}\left((-1)^n - 1\right)$$ You don't need to further simplify

(b) (5 points) On the axes below, sketch the graph of the function to which this series converges on the interval $[-\pi, \pi]$.



5. (25 points) Solve

$$\begin{array}{ll} u_{xx} = 2u_t & 0 < x < 4 \\ u_x(0,t) = 0 & u(4,t) = 0 \\ u_x(x,0) = (4-x)^2 & \end{array}$$

You must show all steps when arriving at your answer (don't jump to form of the final solution). You do NOT need to solve for the coefficients, however you MUST set up their corresponding equation(s).

$$\boxed{u(x,t) = \sum_{n=0}^{\infty} A_n e^{\frac{(2n+1)^2\pi^2}{2(8^2)}t} \cos\left(\frac{(2n+1)\pi x}{8}\right), \quad A_n = \left(\frac{8}{(2n+1)\pi}\right)\left(\frac{1}{2}\right)\int_0^4 (4-x)^2 \sin\left(\frac{(2n+1)\pi x}{8}\right)dx}$$

$$u(x,t) = X(x)T(t)$$

$$\frac{X''}{X} = \frac{2T'}{T} = -\lambda$$

$$X(x) = A\cos\sqrt{\lambda}x + B\sin\sqrt{\lambda}x$$

$$X'(0) = 0 \Rightarrow B = 0$$

$$X(4) = A\cos 4\sqrt{\lambda} = 0$$

$$4\sqrt{\lambda} = \frac{(2n+1)\pi}{2} \quad n = 0, 1, 2, \ldots$$

$$\lambda = \left(\frac{(2n+1)\pi}{8}\right)^2 \quad X_n(x) = \cos\left(\frac{(2n+1)\pi x}{8}\right)$$

$$2T' = -\lambda T$$

$$T' = -\frac{\lambda}{2}T$$

$$T_n(t) = e^{-\frac{(2n+1)^2\pi^2}{2(8^2)}t}$$

$$u(x,t) = \sum_{n=0}^{\infty} A_n e^{-\frac{(2n+1)^2\pi^2}{2(8)^2}t} \cos\left(\frac{(2n+1)\pi x}{8}\right)$$

$$u_x(x,0) = (4-x)^2 = \sum_{n=0}^{\infty} \frac{8}{(2n+1)\pi} A_n \sin\left(\frac{(2n+1)\pi x}{8}\right)$$

6. (20 points) Solve Laplace's equation in polar coordinates

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0,$$

on the disk $0 \leq r \leq 1$, $0 \leq \theta < 2\pi$, with the boundary condition

$$u(1, \theta) = 2 + 3\cos(2\theta) - 4\sin(3\theta).$$

You MUST solve for the coefficients.

$$\boxed{u(r,\theta) = 2 + 3r^2\cos(2\theta) - 4r^3\sin(3\theta)}$$

$u(r,\theta) = R(r)T(\theta)$ $\quad r^2R'' + rR' - n^2R = 0 \qquad T'' + \lambda T = 0$

$T_n(\theta) = A_n\cos(n\theta) + B_n\sin(n\theta)$

$\lambda = n^2,\ n = 0, 1, \ldots$

$$R_n(r) = \begin{cases} \tilde{A}_0 + \tilde{B}_0 \log r & \text{if } n = 0 \\ \tilde{A}_n r^n + \tilde{B}_n r^{-n} & \text{if } n \neq 0 \end{cases}$$

continuous @ $r = 0 \Rightarrow \tilde{B}_0 = \tilde{B}_n = 0$

$$u(r,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} r^n\left(a_n\cos(n\theta) + b_n\sin(n\theta)\right)$$

$$u(1,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(n\theta) + b_n\sin(n\theta) = 2 + 3\cos(2\theta) - 4\sin(3\theta)$$

$a_0 = 4$

$$a_n = \begin{cases} 3 & \text{if } n = 2 \\ 0 & \text{else} \end{cases}$$

$$b_n = \begin{cases} -4 & \text{if } n = 3 \\ 0 & \text{else} \end{cases}$$